Input system:
Dependent variables:x_2, x_3, x_1
Independent variables:t
Equations:

$$\frac{dx_1}{dt} = x_3^3 \sin(\cos(a\ln(x_2)^2 + bx_3)) + x_3^3 b \ln(x_2)^4 + \sin(a\ln(x_2)^2 + bx_3)^5 + \mathrm{Dv}(g^2, 1, (-1); x_1)$$

$$\frac{dx_2}{dt} = x_2^2 \cos(\sin(a\ln(x_2)^2 + bx_3)) + \cos(a\ln(x_2)^2 + bx_3)^4 + \mathrm{EK}(\sin(\sin(a\ln(x_2)^2 + bx_3)), x_1)$$

$$\frac{dx_3}{dt} = x_1 \mathrm{ch}(a\ln(x_2)^2 + bx_3)^2 + x_1 \sin(a\ln(x_2)^2 + bx_3) + \mathrm{sh}(a\ln(x_2)^2 + bx_3)^5$$

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Result:
Dependent variables:$x_2, s_6, s_1, s_{11}, s_{15}, s_{14}, s_8, s_{10}, s_{13}, x_3, x_1, s_{12}, s_3, s_{16}, s_5, s_7, s_9, s_2, s_4$
Independent variables:t
Additional variables:

$$s_1 = \ln(x_2)$$

$$s_2 = \mathrm{inv}(x_2)$$

$$s_3 = \cos(as_1^2 + bx_3)$$

$$s_4 = \sin(as_1^2 + bx_3)$$

$$s_5 = \sin(s_3)$$

$$s_6 = \cos(s_3)$$

$$s_7 = \mathrm{Dv}(g^2, 1, (-1); x_1)$$

$$s_8 = \mathrm{DV}(g^2, 1, (-1); x_1)$$

$$s_9 = \cos(s_4)$$

$$s_{10} = \sin(s_4)$$

$$s_{11} = \mathrm{EK}(s_{10}, x_1)$$

$$s_{12} = \mathrm{EKs}(s_{10}, x_1)$$

$$s_{13} = \mathrm{EKc}(s_{10}, x_1)$$

$$s_{14} = \mathrm{EKi}(s_{10}, x_1)$$

$$s_{15} = \mathrm{ch}(as_1^2 + bx_3)$$

$$s_{16} = \mathrm{sh}(as_1^2 + bx_3)$$

Equations:

$$\frac{dx_1}{dt} = x_3^3 b s_1^4 + s_4^5 + x_3^3 s_5 + s_7$$

$$\frac{dx_2}{dt} = s_3^4 + x_2^2 s_9 + s_{11}$$

$$\frac{dx_3}{dt} = s_{16}^5 + x_1 s_{15}^2 + x_1 s_4$$

$$\frac{ds_1}{dt} = s_2 s_3^4 + s_2 x_2^2 s_9 + s_2 s_{11}$$

$$\frac{ds_2}{dt} = (-1) s_2^2 s_3^4 + (-1) s_2^2 x_2^2 s_9 + (-1) s_2^2 s_{11}$$

$$\begin{aligned}\frac{ds_3}{dt} = {} & (-2) s_4 a s_1 s_2 s_3^4 + (-2) s_4 a s_1 s_2 x_2^2 s_9 + (-1) s_4 b s_{16}^5 \\ & + (-2) s_4 a s_1 s_2 s_{11} + (-1) s_4 b x_1 s_{15}^2 + (-1) s_4^2 b x_1\end{aligned}$$

$$\frac{ds_4}{dt} = 2 s_3^5 a s_1 s_2 + 2 s_3 a s_1 s_2 x_2^2 s_9 + s_3 b s_{16}^5 + 2 s_3 a s_1 s_2 s_{11} + s_3 b x_1 s_{15}^2 + s_3 b x_1 s_4$$

$$\begin{aligned}\frac{ds_5}{dt} = {} & (-2) s_6 s_4 a s_1 s_2 s_3^4 + (-2) s_6 s_4 a s_1 s_2 x_2^2 s_9 + (-1) s_6 s_4 b s_{16}^5 \\ & + (-2) s_6 s_4 a s_1 s_2 s_{11} + (-1) s_6 s_4 b x_1 s_{15}^2 + (-1) s_6 s_4^2 b x_1\end{aligned}$$

$$\begin{aligned}\frac{ds_6}{dt} = {} & 2 s_5 s_4 a s_1 s_2 s_3^4 + 2 s_5 s_4 a s_1 s_2 x_2^2 s_9 + s_5 s_4 b s_{16}^5 \\ & + 2 s_5 s_4 a s_1 s_2 s_{11} + s_5 s_4 b x_1 s_{15}^2 + s_5 s_4^2 b x_1\end{aligned}$$

$$\frac{ds_7}{dt} = s_8 x_3^3 b s_1^4 + s_8 s_4^5 + s_8 x_3^3 s_5 + s_8 s_7$$

$$\begin{aligned}\frac{ds_8}{dt} = {} & (-1) x_3^3 b s_1^4 s_7 p_1 x_1^2 + (-1) x_3^3 b s_1^4 s_7 p_2 x_1 + (-1) x_3^3 b s_1^4 s_7 p_3 + (-1) s_4^5 s_7 p_1 x_1^2 \\ & + (-1) x_3^3 s_5 s_7 p_1 x_1^2 + (-1) s_4^5 s_7 p_2 x_1 + (-1) x_3^3 s_5 s_7 p_2 x_1 + (-1) s_4^5 s_7 p_3 \\ & + (-1) x_3^3 s_5 s_7 p_3 + (-1) s_7^2 p_1 x_1^2 + (-1) s_7^2 p_2 x_1 + (-1) s_7^2 p_3\end{aligned}$$

$$\frac{ds_9}{dt} = (-2)s_{10}s_3^5as_1s_2 + (-2)s_{10}s_3as_1s_2x_2^2s_9 + (-1)s_{10}s_3bs_{16}^5 + (-2)s_{10}s_3as_1s_2s_{11} + (-1)s_{10}s_3bx_1s_{15}^2 + (-1)s_{10}s_3bx_1s_4$$

$$\frac{ds_{10}}{dt} = 2s_9s_3^5as_1s_2 + 2s_9^2s_3as_1s_2x_2^2 + s_9s_3bs_{16}^5 + 2s_9s_3as_1s_2s_{11} + s_9s_3bx_1s_{15}^2 + s_9s_3bx_1s_4$$

$$\frac{ds_{11}}{dt} = 2s_{12}s_{14}s_9s_3^5as_1s_2 + 2s_{12}s_{14}s_9^2s_3as_1s_2x_2^2 + s_{12}s_{14}s_9s_3bs_{16}^5 + s_{14}x_3^3bs_1^4 + 2s_{12}s_{14}s_9s_3as_1s_2s_{11} + s_{12}s_{14}s_9s_3bx_1s_{15}^2 + s_{12}s_{14}s_9s_3bx_1s_4 + s_{14}s_4^5 + s_{14}x_3^3s_5 + s_{14}s_7$$

$$\frac{ds_{12}}{dt} = 2s_{12}s_{13}s_{14}s_9s_3^5as_1s_2 + 2s_{12}s_{13}s_{14}s_9^2s_3as_1s_2x_2^2 + s_{12}s_{13}s_{14}s_9s_3bs_{16}^5 + s_{13}s_{14}x_3^3bs_1^4 + 2s_{12}s_{13}s_{14}s_9s_3as_1s_2s_{11} + s_{12}s_{13}s_{14}s_9s_3bx_1s_{15}^2 + s_{12}s_{13}s_{14}s_9s_3bx_1s_4 + s_{13}s_{14}s_4^5 + s_{13}s_{14}x_3^3s_5 + s_{13}s_{14}s_7$$

$$\frac{ds_{13}}{dt} = (-2)s_{12}^2s_{14}s_9s_3^5as_1s_2 + (-2)s_{12}^2s_{14}s_9^2s_3as_1s_2x_2^2 + (-1)s_{12}^2s_{14}s_9s_3bs_{16}^5 + (-1)s_{12}s_{14}x_3^3bs_1^4 + (-2)s_{12}^2s_{14}s_9s_3as_1s_2s_{11} + (-1)s_{12}^2s_{14}s_9s_3bx_1s_{15}^2 + (-1)s_{12}^2s_{14}s_9s_3bx_1s_4 + (-1)s_{12}s_{14}s_4^5 + (-1)s_{12}s_{14}x_3^3s_5 + (-1)s_{12}s_{14}s_7$$

$$\frac{ds_{14}}{dt} = (-2)s_{10}s_{12}^2s_{14}^3s_9s_3^5as_1s_2 + (-2)s_{10}s_{12}^2s_{14}^3s_9^2s_3as_1s_2x_2^2 + (-1)s_{10}s_{12}^2s_{14}^3s_9s_3bs_{16}^5 + (-1)s_{10}s_{14}^3s_{12}x_3^3bs_1^4 + 2s_{13}s_{14}^2s_9s_3^5as_1s_2 + (-2)s_{10}s_{12}^2s_{14}^3s_9s_3as_1s_2s_{11} + (-1)s_{10}s_{12}^2s_{14}^3s_9s_3bx_1s_{15}^2 + 2s_{13}s_{14}^2s_9^2s_3as_1s_2x_2^2 + s_{13}s_{14}^2s_9s_3bs_{16}^5 + (-1)s_{10}s_{12}^2s_{14}^3s_9s_3bx_1s_4 + (-1)s_{10}s_{14}^3s_{12}s_4^5 + 2s_{13}s_{14}^2s_9s_3as_1s_2s_{11} + s_{13}s_{14}^2s_9s_3bx_1s_{15}^2 + (-1)s_{10}s_{14}^3s_{12}x_3^3s_5 + s_{13}s_{14}^2s_9s_3bx_1s_4 + (-1)s_{10}s_{14}^3s_{12}s_7$$

$$\frac{ds_{15}}{dt} = 2s_{16}as_1s_2s_3^4 + 2s_{16}as_1s_2x_2^2s_9 + s_{16}^6b + 2s_{16}as_1s_2s_{11} + s_{16}bx_1s_{15}^2 + s_{16}bx_1s_4$$

$$\frac{ds_{16}}{dt} = 2s_{15}as_1s_2s_3^4 + 2s_{15}as_1s_2x_2^2s_9 + s_{15}bs_{16}^5 + 2s_{15}as_1s_2s_{11} + s_{15}^3bx_1 + s_{15}bx_1s_4$$